Performance Metrics for Binary Classification

Lecture Notes with Teaching Strategies

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Week 6: Performance Metrics for Binary Classification

Topics Covered: Confusion Matrix, Accuracy, Precision, Recall, F1-Score, ROC Curve, AUC, Domain-Specific Applications, Imbalanced Data Handling

Contents

Learning Objectives

By the end of this lecture, you will be able to:

- Construct and interpret confusion matrices for binary classification
- Calculate and interpret accuracy, precision, recall, and F1-score
- Understand when to use different metrics based on application context
- Plot and interpret ROC curves
- Calculate AUC and understand its meaning
- Handle imbalanced datasets using appropriate metrics
- Apply performance metrics to real-world ML problems

Theoretical Foundations

Core Concepts:

- **Binary Classification:** Predicting one of two possible outcomes (0/1, Yes/No, Positive/Negative)
- **Confusion Matrix:** 2×2 table showing TP, FP, TN, FN
- **Type I vs Type II Errors:** False Positive vs False Negative
- **Precision-Recall Tradeoff:** Relationship between these metrics
- **ROC Curve:** Visual representation of classifier performance
- **AUC:** Area Under ROC Curve - single performance metric
- **Threshold Selection:** Balancing different error types

1 Introduction to Performance Metrics

1.1 Why Performance Metrics Matter

After training a machine learning model, we need to evaluate its performance quantitatively. Different metrics answer different questions: - How often is the model correct overall? (Accuracy) - When it predicts positive, how often is it correct? (Precision) - How many actual positives does it catch? (Recall) - How well does it distinguish between classes? (AUC)

1.2 The Confusion Matrix: Foundation of All Metrics

Definition

Confusion Matrix: A 2×2 table that compares actual vs predicted classes in binary classification. It provides complete information about classification performance.

Confusion Matrix Structure

		Actual Class	
		Positive (1)	**Negative (0)**
2*Predicted	**Positive (1)**	**TP**	**FP**
	Negative (0)	**FN**	**TN**

1.2.1 Terminology Explained

- **True Positive (TP):** Model correctly predicts positive
- **False Positive (FP):** Model incorrectly predicts positive (Type I Error)
- **True Negative (TN):** Model correctly predicts negative
- **False Negative (FN):** Model incorrectly predicts negative (Type II Error)

1.2.2 Step-by-Step Construction

Worked Example

Example: Building a Confusion Matrix
Given actual labels y and predicted labels $\hat{y}$:

Data Point	Actual (y)	Predicted ($\hat{y}$)
1	0	1
2	1	1
3	0	0
4	1	1
5	1	1
6	0	1
7	1	0

Solution:
Count each combination:

- **TP**: Actual=1, Predicted=1 $\rightarrow$ Points 2, 4, 5 $\rightarrow$ 3
- **FP**: Actual=0, Predicted=1 $\rightarrow$ Points 1, 6 $\rightarrow$ 2
- **TN**: Actual=0, Predicted=0 $\rightarrow$ Point 3 $\rightarrow$ 1
- **FN**: Actual=1, Predicted=0 $\rightarrow$ Point 7 $\rightarrow$ 1

Confusion Matrix:

	Actual=1	Actual=0
Predicted=1	**TP=3**	**FP=2**
Predicted=0	**FN=1**	**TN=1**

2 Basic Performance Metrics

2.1 Accuracy

Definition

Accuracy: The ratio of correctly predicted observations to total observations.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

From our example:

$$\text{Accuracy} = \frac{3+1}{3+1+2+1} = \frac{4}{7} \approx 0.571 \text{ or } 57.1\%$$

2.1.1 Limitations of Accuracy

Problem with Imbalanced Datasets: Consider a dataset with 990 negative and 10 positive samples. A naive model that always predicts negative would have:

$$\text{Accuracy} = \frac{0+990}{1000} = 99\%$$

But it completely fails to identify any positive cases!

2.2 Precision

Definition

Precision: Measures how many of the predicted positives are actually positive. Answers: "When we predict positive, how often are we right?"

$$\text{Precision} = \frac{TP}{TP + FP}$$

From our example:

$$\text{Precision} = \frac{3}{3+2} = \frac{3}{5} = 0.6 \text{ or } 60\%$$

2.3 Recall (Sensitivity)

Definition

Recall: Measures how many of the actual positives are correctly predicted. Answers: "Of all actual positives, how many did we catch?"

$$\text{Recall} = \frac{TP}{TP + FN}$$

Also called Sensitivity or True Positive Rate (TPR).

From our example:

$$\text{Recall} = \frac{3}{3+1} = \frac{3}{4} = 0.75 \text{ or } 75\%$$

2.4 F1-Score

Definition

F1-Score: Harmonic mean of precision and recall. Balances both metrics.

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

From our example:

$$F_1 = 2 \times \frac{0.6 \times 0.75}{0.6 + 0.75} = 2 \times \frac{0.45}{1.35} = 0.667$$

2.5 Comprehensive Example

Worked Example

Example: Medical Diagnosis System
A cancer detection model was tested on 1000 patients: - TP = 85 (correctly identified cancer) - FP = 15 (healthy patients flagged as having cancer) - TN = 880 (correctly identified healthy) - FN = 20 (missed cancer cases)
Calculate all metrics.
Solution:
1. Confusion Matrix:

	Actual Cancer	Actual Healthy
Predicted Cancer	85	15
Predicted Healthy	20	880

2. Metrics:

$$\text{Accuracy} = \frac{85 + 880}{1000} = 0.965\ (96.5\%)$$

$$\text{Precision} = \frac{85}{85 + 15} = 0.85\ (85\%)$$

$$\text{Recall} = \frac{85}{85 + 20} = 0.8095\ (80.95\%)$$

$$F_1 = 2 \times \frac{0.85 \times 0.8095}{0.85 + 0.8095} = 0.829\ (82.9\%)$$

3 Advanced Metrics: ROC and AUC

3.1 ROC Curve (Receiver Operating Characteristic)

Definition

ROC Curve: A plot of True Positive Rate (Recall) vs False Positive Rate at various classification thresholds.

$$\text{TPR} = \frac{TP}{TP + FN} \quad \text{(Recall)}$$

$$\text{FPR} = \frac{FP}{FP + TN}$$

3.1.1 Interpreting ROC Curves

- **Perfect Classifier:** Top-left corner (TPR=1, FPR=0)
- **Random Classifier:** Diagonal line (TPR = FPR)
- **Good Classifier:** Curve above diagonal, closer to top-left
- **Changing Threshold:** Moves along the curve

3.2 AUC (Area Under ROC Curve)

Definition

AUC: Area Under the ROC Curve. Ranges from 0 to 1. Provides a single metric for classifier performance across all thresholds.

- AUC = 0.5: Random classifier
- AUC = 1.0: Perfect classifier
- AUC > 0.8: Good classifier
- AUC < 0.5: Worse than random (possibly inverted labels)

3.3 Example: ROC Curve Calculation

Worked Example

Example: Calculating TPR and FPR
Given a classifier with probabilities and different thresholds:

Sample	Actual	Probability	Predicted (thresh=0.5)
1	1	0.9	1
2	0	0.8	1
3	1	0.7	1
4	0	0.6	1
5	1	0.4	0
6	0	0.3	0

Solution:
At threshold = 0.5: - TP = 2 (samples 1,3) - FP = 2 (samples 2,4) - TN = 1 (sample 6) - FN = 1 (sample 5)

$$\text{TPR} = \frac{2}{2+1} = 0.667$$

$$\text{FPR} = \frac{2}{2+1} = 0.667$$

This gives point (0.667, 0.667) on ROC curve.

4 Domain-Specific Applications

Practical Applications in Data Science & ML

Practical Applications in Data Science & ML:

- **Medical Diagnosis:** High recall crucial (minimize FN)
- **Fraud Detection:** High precision important (minimize FP)
- **Spam Filtering:** High precision critical (avoid losing important emails)
- **Recommendation Systems:** Balanced precision and recall
- **Credit Scoring:** High precision for loan approval
- **Manufacturing QC:** High recall for defect detection
- **Security Systems:** Low FPR to avoid false alarms
- **Marketing Campaigns:** High precision for customer targeting
- **Legal Document Review:** High recall to avoid missing relevant documents
- **Autonomous Vehicles:** Extremely high precision for object detection

4.1 Domain-Specific Metric Selection

5 Handling Imbalanced Data

Application	Key Metric	Reason
Medical Diagnosis	Recall	Missing disease is catastrophic
Spam Filtering	Precision	Important emails in spam folder is unacceptable
Fraud Detection	Precision & Recall	Both errors costly (false alarms vs missed fraud)
Search Engines	Precision	Top results must be relevant
Anomaly Detection	Recall	Catching all anomalies is critical

Table 1: Metric selection based on application

5.1 Common Techniques

- **Resampling:**
  - Oversampling minority class
  - Undersampling majority class
  - SMOTE (Synthetic Minority Oversampling)
- **Cost-Sensitive Learning:**
  - Assign higher cost to misclassifying minority class
  - Weighted loss functions
- **Alternative Metrics:**
  - Precision-Recall curves (better for imbalanced data)
  - Matthews Correlation Coefficient (MCC)
  - Cohen's Kappa

5.2 Precision-Recall Curve

For highly imbalanced datasets, Precision-Recall (PR) curves are often more informative than ROC curves. The baseline is the fraction of positives in the dataset.

6 Practice Problems

6.1 Problem 1: Basic Metrics Calculation

A binary classifier on test data of 200 samples gives: - TP = 40, FP = 10, TN = 140, FN = 10
Calculate: Accuracy, Precision, Recall, F1-Score

6.2 Problem 2: Confusion Matrix Construction

Given: - Total samples: 500 - Accuracy: 88% - Recall: 70% - Precision: 60% - Number of actual positives: 100
Find the complete confusion matrix (TP, FP, TN, FN).

6.3 Problem 3: Domain Analysis

You're building:

1. A system to detect defective products in manufacturing
2. A system to filter spam emails
3. A system for credit card fraud detection

For each system, which metric is most important and why?

6.4 Problem 4: ROC Curve Analysis

A classifier has the following performance at different thresholds:

Threshold	TPR	FPR
0.9	0.2	0.05
0.7	0.6	0.1
0.5	0.8	0.3
0.3	0.95	0.5
0.1	0.99	0.8

Plot the ROC curve and estimate AUC.

6.5 Problem 5: Imbalanced Data

In a dataset of 10,000 transactions: - 9,900 legitimate (negative class) - 100 fraudulent (positive class)

A model predicts all transactions as legitimate. Calculate accuracy and explain why this metric is misleading.

7 Solutions to Practice Problems

7.0.1 Solution 1: Basic Metrics

$$\text{Accuracy} = \frac{40 + 140}{200} = 0.9\ (90\%)$$

$$\text{Precision} = \frac{40}{40 + 10} = 0.8\ (80\%)$$

$$\text{Recall} = \frac{40}{40 + 10} = 0.8\ (80\%)$$

$$F_1 = 2 \times \frac{0.8 \times 0.8}{0.8 + 0.8} = 0.8\ (80\%)$$

7.0.2 Solution 2: Confusion Matrix

Given: - Total = 500 - Actual Positives = 100 → Actual Negatives = 400 - Recall = 0.7 = TP/100 → TP = 70 - FN = 100 - 70 = 30 - Precision = 0.6 = 70/(70 + FP) → FP = 70/0.6 - 70 = 116.67 - 70 47 - Accuracy = 0.88 = (TP + TN)/500 → TN = 500*0.88 - 70 = 440 - 70 = 370

Check: TP + FP + TN + FN = 70 + 47 + 370 + 30 = 517 500 Adjust: Let's solve systematically...

7.0.3 Solution 3: Domain Analysis

1. **Defect Detection:** Recall most important (minimize FN)
2. **Spam Filtering:** Precision most important (minimize FP)
3. **Fraud Detection:** Both important, but often precision prioritized to avoid false alarms

7.0.4 Solution 4: ROC Curve

Points: (0.05, 0.2), (0.1, 0.6), (0.3, 0.8), (0.5, 0.95), (0.8, 0.99) AUC 0.85 (estimate by trapezoidal rule)

7.0.5 Solution 5: Imbalanced Data

Model accuracy = 9900/10000 = 99% This is misleading because it never detects any fraudulent transactions (Recall = 0%, Precision = undefined).

8 Summary and Key Takeaways

- **Confusion Matrix** is fundamental - know TP, FP, TN, FN
- **Accuracy** is simple but misleading for imbalanced data
- **Precision** focuses on prediction quality (avoid FP)
- **Recall** focuses on finding all positives (avoid FN)
- **F1-Score** balances precision and recall
- **ROC Curve** shows tradeoff between TPR and FPR
- **AUC** provides single performance metric
- **Metric selection** depends on application context
- For **imbalanced data**, use appropriate techniques and metrics

Learning Resources

Recommended Resources:

- **Textbook:** "Introduction to Statistical Learning" (James et al.)
- **Online Course:** Andrew Ng's Machine Learning on Coursera
- **Python Libraries:** scikit-learn (metrics module)
- **Interactive Tutorials:** Kaggle courses on model evaluation
- **Research Paper:** "The Relationship Between Precision-Recall and ROC Curves" by Davis & Goadrich

Next Week Preview

Next Week Preview: Multiclass Classification Metrics

- Extension of binary metrics to multiple classes
- Macro, Micro, and Weighted averages
- Confusion matrices for multiclass problems
- One-vs-Rest and One-vs-One strategies
- Application to text classification and image recognition